Nandha Exam & Result Portal

Project Requirements & Architectural Specification Blueprint

1. Executive Summary

Nandha Exam & Result Portal is designed as an intelligent, automated digital platform specifically tailored to solve manual workflow friction for academic. The platform centralizes data operations, eliminates redundant paperwork, and enforces data accuracy and compliance with sub-second responsiveness.

2. Problem Statement

Currently, stakeholders rely on disjointed spreadsheets, manual data compilation, and error-prone physical paperwork. This leads to operational bottlenecks, delayed reports, and an estimated 10-15 wasted hours per staff member every week.

3. Project Goals

- Automate end-to-end data processing for Nandha Exam & Result Portal
- Enable high-velocity document export in compliant PDF, Word, and Excel formats
- Provide auditable, role-based security ensuring sensitive student and organizational records remain protected
- Reduce report compilation latency from days to under 10 seconds

4. Confirmed Requirements

Requirement	Category	Priority	MVP
AI Academic Workspace (MCQs & Question Paper Generator)	Functional	Must Have	Yes
Document & Excel Automation (Bulk Marks & Calculations)	Functional	Must Have	Yes
Result Analysis & Visual Insights	Functional	Must Have	Yes

5. Page-by-Page UI Architecture

Login & Onboarding — *Route: /login*

Key Components: Email/password form, SSO button, Password recovery, Role preview

Executive Dashboard — *Route: /dashboard*

Key Components: Recent activities, quick action launchpad, system status cards, pending approvals

Document & Academic Workspace — *Route: /workspace*

Key Components: File upload dropzone, live template previewer, export toolbar (PDF/DOCX/Excel)

Analytics & Results Center — *Route: /analytics*

Key Components: Interactive charts, pass percentage distribution, arrear identification table

Administration & Settings — *Route: /admin*

Key Components: User role management, question templates editor, security & audit trail viewer

6. Security, Compliance & Governance

- Password hashing using industry-standard bcrypt with salt
- JWT tokens signed with HS256 and configurable expiration
- Server-side authorization guards preventing unauthorized tenant access
- Sanitization of all incoming filenames and uploaded document streams
- Audit trail tracking all export and project administrative operations

7. Acceptance Criteria

- [ ] User can input a project idea in English or Tamil-English transliteration and have it auto-structured
- [ ] Questions adapt in real time to previous selections without page reloads
- [ ] Irrelevant questions are automatically omitted from the active flow
- [ ] Auto-save preserves every answer with visual confirmation
- [ ] Clicking export generates real downloadable PDF, DOCX, Markdown, JSON, and CSV files
- [ ] The entire platform runs responsively across mobile and desktop viewports